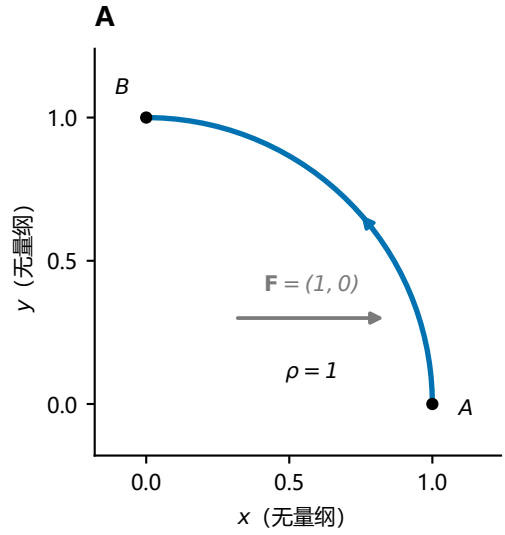
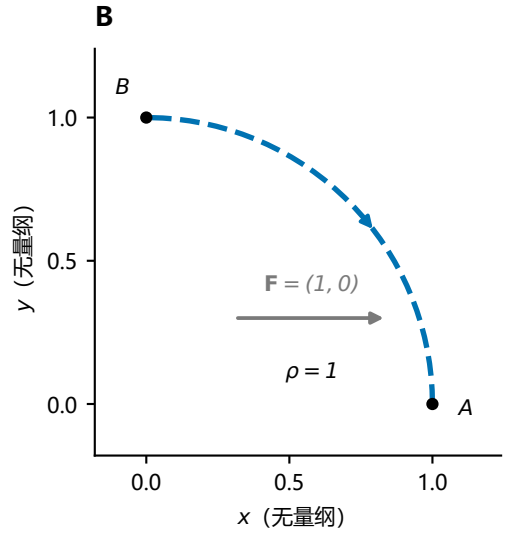


$C : A \to B$

$\int_C \mathrm{d}s = \pi/2, \quad \int_C \mathrm{d}x = -1$



$-C : B \to A$

$\int_{-C} \mathrm{d}s = \pi/2, \quad \int_{-C} \mathrm{d}x = 1$